

ABDULLA AL MAMUN

Ph.D. (Hydrology & Water Resources) | M.Sc. Mathematics (×2)

Postdoctoral Fellow, Hohai University, Nanjing, China

Email: abdullamamun21@gmail.com

Mobile: +880 17411 83886 | +86 15195963853

Google Scholar: <https://scholar.google.com/citations?user=2qWex3oAAAAJ>

ORCID: <https://orcid.org/0000-0003-0183-7112>

RESEARCH PROFILE

Applied mathematician specializing in fractional partial differential equations (PDEs), nonlinear wave phenomena, and computational modeling of fluid dynamics and hydrological systems. Strong record of peer-reviewed publications in high-impact journals (e.g., *Scientific Reports*, *Nonlinear Dynamics*, *Chaos*, *Solitons & Fractals*). My work integrates analytical techniques with high-performance numerical simulation to study soliton dynamics, bifurcation, and stability in complex systems. Active reviewer and **Reviewing Editor for Springer Nature**.

RESEARCH INTERESTS

- Applied & Computational Mathematics
- Linear and Nonlinear Partial Differential Equations
- Fractional Calculus and Fractional PDEs
- Numerical Analysis and Scientific Computing
- Dynamical Systems and Soliton Theory
- Mathematical Modeling of Fluid Dynamics and Water Waves

EDUCATION

- Ph.D. in Hydrology and Water Resources — Hohai University, Nanjing, China (2025)
- M.Sc. in Mathematics — Hohai University, Nanjing, China (2021)
- M.Sc. in Mathematics — Islamic University, Kushtia, Bangladesh (2019)
- B.Sc. (Hons) in Mathematics — Islamic University, Kushtia, Bangladesh (2017)

ACADEMIC APPOINTMENTS

- **Postdoctoral Fellow** — School of Earth Science and Engineering, Hohai University, Nanjing, China (2025–Present)
- **Lecturer in Mathematics** — Department of Computer Science and Engineering, Northern University of Business and Technology Khulna (NUBTK), Khulna, Bangladesh (2022–2025)
- **Lecturer in Mathematics** — Department of Information and Communication Engineering, Rabindra Maitree University, Kushtia, Bangladesh (2021–2022)

- **Lecturer in Mathematics** — Department of Computer Science and Engineering, Institute of Science Trade and Technology (ISTT), Dhaka, Bangladesh (2017–2018)

PUBLICATIONS

1. **Mamun, A. A.**, Ananna, S. N., Lu, C., Zhang, Z., Biswas, M., Limon, S. B. A., & Asaduzzaman, M. (2025). Impact of numerical simulation and fractionality of the 3D fractional WBBM model using GERF and MGERF methods. *Alexandria Engineering Journal*, 134, 621–643. <https://doi.org/10.1016/j.aej.2025.12.043>
2. Zhang, Z., An, T., Bu, W., Li, S., & **Mamun, A. A.** (2025). Existence results for Robin problems involving $p(x)$ -Laplacian-like operators with convection term. *Demonstratio Mathematica*, 58(1), 20250180. <https://doi.org/10.1515/dema-2025-0180>
3. Ananna, S. N., **Mamun, A. A.**, Meia, M. A. A., & Lu, C. (2025). Chaotic behaviour, bifurcation, and stability analysis of the time-fractional phi-four model using the Hirota bilinear form. *International Journal of Computer Mathematics*, 1–21. <https://doi.org/10.1080/00207160.2025.2538525>
4. **Mamun, A. A.**, Ananna, S. N., & Lu, C. (2025). Numerical investigation of bifurcation soliton, lump, and interaction solutions for the 3D fractional WBBM model using the Hirota bilinear form. *Nonlinear Dynamics*, 113, 13631–13648. <https://doi.org/10.1007/s11071-025-11043-0>
5. Uddin, M. M., Fang, G., Huang, X., Gordillo, J. R. I., Tapu, M. a. I., & **Mamun, A. A.** (2024). The ecological operation of ChiTan hydropower station based on hydrological alteration using PCA method. *Hydrology Research*, 55(11), 1069–1090. <https://doi.org/10.2166/nh.2024.013>
6. Asaduzzaman, M., Kilicman, A., **Al-Mamun, A.**, & Hossain, M. D. (2024). Analysis of a novel conformable fractional order ASIR dengue transmission model in the perspective of Bangladesh. *Mathematical Models and Computer Simulations*, 16(3), 431–456. <https://doi.org/10.1134/s2070048224700157>
7. **Mamun, A. A.**, Lu, C., Ananna, S. N., Ismail, H. G., Bari, M. A., & Uddin, M. M. (2024). The influence of fractionality and unconstrained parameters on mathematical and graphical analysis of the time fractional phi-four model. *Chaos, Solitons & Fractals*, 183, 114892. <https://doi.org/10.1016/j.chaos.2024.114892>
8. **Mamun, A. A.**, Lu, C., Ananna, S. N., & Uddin, M. M. (2024). Rational Sine-Gordon expansion method to analyze the dynamical behavior of the time-fractional phi-four and (2+1) dimensional CBS equations. *Scientific Reports*, 14, 9473. <https://doi.org/10.1038/s41598-024-60156-w>
9. **Mamun, A. A.**, Lu, C., Ananna, S. N., & Uddin, M. M. (2024). Dynamical behavior of water wave phenomena for the 3D fractional WBBM equations using rational sine-Gordon expansion method. *Scientific Reports*, 14, 6455. <https://doi.org/10.1038/s41598-024-55215-1>
10. **Mamun, A. A.**, Ananna, S. N., Gharami, P. P., An, T., Liu, W., & Asaduzzaman, M. (2023). An innovative approach for developing the precise traveling wave solutions to a family of 3D fractional WBBM equations. *Partial Differential Equations in Applied Mathematics*, 100522. <https://doi.org/10.1016/j.padiff.2023.100522>

11. Gharami, P. P., **Mamun, A. A.**, Gazi, M. A., Ananna, S. N., & Ahmmed, S. F. (2022). Numerical exploration of MHD unsteady flow of THNF passing through a moving cylinder with Soret and Dufour effects. *Partial Differential Equations in Applied Mathematics*, 100463. <https://doi.org/10.1016/j.padiff.2022.100463>
12. **Mamun, A. A.**, Ananna, S. N., Gharami, P. P., An, T., & Asaduzzaman, M. (2022). The improved modified extended tanh-function method to develop the exact travelling wave solutions of a family of 3D fractional WBBM equations. *Results in Physics*, 41, 105969. <https://doi.org/10.1016/j.rinp.2022.105969>
13. **Mamun, A. A.**, Ananna, S. N., An, T., Asaduzzaman, M., & Rana, M. S. (2022). Sine-Gordon expansion method to construct the solitary wave solutions of a family of 3D fractional WBBM equations. *Results in Physics*, 40, 105845. <https://doi.org/10.1016/j.rinp.2022.105845>
14. **Mamun, A. A.**, Ananna, S. N., An, T., Asaduzzaman, M., & Hasan, A. (2021). Optical soliton analysis to a family of 3D WBBM equations with conformable derivative via a dynamical approach. *Partial Differential Equations in Applied Mathematics*, 100238. <https://doi.org/10.1016/j.padiff.2021.100238>
15. **Mamun, A. A.**, Ananna, S. N., An, T., Asaduzzaman, M., & Miah, M. M. (2021). Solitary wave structures of a family of 3D fractional WBBM equation via the tanh-coth approach. *Partial Differential Equations in Applied Mathematics*, 100237. <https://doi.org/10.1016/j.padiff.2021.100237>
16. Ananna, S. N., **Mamun, A. A.**, & An, T. Q. (2021). Periodic wave analysis to the time-fractional phi-four and (2+1) dimensional CBS equations. *International Journal of Physical Research*, 9(2), 98–104. <https://doi.org/10.14419/ijpr.v9i2.31824>
17. Alam, L. M. B., Xingfang, J., **Mamun, A. A.**, & Ananna, S. N. (2021). Investigation of lump, soliton, periodic, kink, and rogue waves to the time-fractional phi-four and (2+1) dimensional CBS equations in mathematical physics. *Partial Differential Equations in Applied Mathematics*, 100122. <https://doi.org/10.1016/j.padiff.2021.100122>
18. **Mamun, A. A.**, Ananna, S. N., An, T., Shahren, N. H. M., Asaduzzaman, M., & Foyjonnesa. (2021). Dynamical behaviour of travelling wave solutions to the conformable time-fractional modified Liouville and mRLW equations in water wave mechanics. *Heliyon*, 7(8), e07704. <https://doi.org/10.1016/j.heliyon.2021.e07704>
19. **Mamun, A. A.**, Shahren, N. H. M., Ananna, S. N., Asaduzzaman, M., & Foyjonnesa. (2021). Solitary and periodic wave solutions to the family of new 3D fractional WBBM equations in mathematical physics. *Heliyon*, 7(7), e07483. <https://doi.org/10.1016/j.heliyon.2021.e07483>
20. Alam, L. M. B., Jiang, X., & **Mamun, A. A.** (2021). Exact and explicit travelling wave solution to the time-fractional phi-four and (2+1) dimensional CBS equations using the modified extended tanh-function method in mathematical physics. *Partial Differential Equations in Applied Mathematics*, 100039. <https://doi.org/10.1016/j.padiff.2021.100039>
21. Shahren, N. H. M., Foyjonnesa, Ali, M. S., **Mamun, A. A.**, & Rahman, M. (2021). Interaction among lump, periodic, and kink solutions with dynamical analysis to the conformable

- time-fractional Phi-four equation. *Partial Differential Equations in Applied Mathematics*, 100038. <https://doi.org/10.1016/j.padiff.2021.100038>
22. **Mamun, A. A.**, Ananna, S. N., An, T., Shahen, N. H. M., & Foyjonnesa. (2021). Periodic and solitary wave solutions to a family of new 3D fractional WBBM equations using the two-variable method. *Partial Differential Equations in Applied Mathematics*, 3, 100033. <https://doi.org/10.1016/j.padiff.2021.100033>
 23. **Mamun, A. A.**, An, T., Shahen, N. H. M., Ananna, S. N., Foyjonnesa, Hossain, M. F., & Muazu, T. (2020). Exact and explicit travelling-wave solutions to the family of new 3D fractional WBBM equations in mathematical physics. *Results in Physics*, 19, 103517. <https://doi.org/10.1016/j.rinp.2020.103517>
 24. Shahen, N. H. M., Foyjonnesa, Bashar, M. H., Ali, M. S., & **Mamun, A. A.** (2020). Dynamical analysis of long-wave phenomena for the nonlinear conformable space-time fractional (2+1)-dimensional AKNS equation in water wave mechanics. *Heliyon*, 6(10), e05276. <https://doi.org/10.1016/j.heliyon.2020.e05276>
 25. Ananna, S. N., & **Mamun, A. A.** (2020). Solution of Volterra's integro-differential equations by using variational iteration method. *International Journal of Scientific & Engineering Research*, 11(3), 1–9.
 26. Ahmed, M. F., & **Mamun, A. A.** (2019). Design of upper and lower material manipulators. *International Journal of Modern Physics and Applications*, 5(2), 38–49.
 27. **Mamun, A. A.**, Asaduzzaman, M., & Ananna, S. N. (2019). Solution of eighth order boundary value problem by using variational iteration method. *International Journal of Mathematics and Computational Science*, 5(2), 13–23.
 28. **Mamun, A. A.**, & Asaduzzaman, M. (2019). Solution of seventh order boundary value problem by using variational iteration method. *International Journal of Mathematics and Computational Science*, 5(1), 6–12.
 29. **Mamun, A. A.**, Ali, M. S., & Miah, M. M. (2018). A study on an analytic solution to the 1D heat equation of a parabolic partial differential equation and its implementation in computer programming. *International Journal of Scientific & Engineering Research*, 9(9), 913–921.

ACADEMIC SERVICE & EDITORIAL ROLES

- **Reviewing Editor**, Springer Nature (March 2026 – Present)
- Reviewed manuscripts for:
 - **Elsevier**: *Journal of Ocean Engineering and Science*, *Results in Physics*, *Fundamental Plasma Physics*
 - **Springer**: *Nonlinear Dynamics*, *Scientific Reports*, *Optical and Quantum Electronics*, *Qualitative Theory of Dynamical Systems*, *Journal of the Korean Physical Society*, *Zeitschrift für angewandte Mathematik und Physik*
 - **MDPI**: *Mathematics*, *Symmetry*, *Fractal & Fractional*
 - **Hindawi**, **World Scientific**, **AIMS Press**, and others.

TEACHING EXPERIENCE

Courses Taught (Undergraduate level):

- Differential & Integral Calculus
- Vector Analysis
- ODEs & PDEs
- Linear Algebra
- Numerical Methods
- Complex Variables
- Fourier & Laplace Analysis
- Discrete Mathematics

GRANTS, SCHOLARSHIPS & AWARDS

- Top 10 Outstanding Student Award — Hohai University (2024)
- Jiangsu Government Scholarship awards for outstanding academic performance and excellence in research (2023, 2024)
- IAHR-Hohai Ph.D. Scholarship (2021)
- Hohai University 1st class Scholarship (2019)
- Merit Scholarships — Islamic University (2013–2015)

LANGUAGES

- Bengali — Native
- English — Professional Working Proficiency
- Chinese — HSK-III Certification

COMPUTER SKILLS

- MATLAB, Maple, Mathematica
- Python, C/C++, Fortran
- Ansys, Abaqus
- MS Office Suite

REFERENCES

- **Prof. Chunhui Lu** — *Ph.D. Supervisor*
College of Hydrology and Water Resources, Hohai University, Nanjing, China
Email: clu@hhu.edu.cn
- **Prof. Tianqing An** — *M.Sc. Supervisor*
School of Mathematics, Hohai University, Nanjing, China
Email: antq@hhu.edu.cn
- **Prof. Md. Asaduzzaman** — *Undergraduate & Graduate Advisor*
Department of Mathematics, Islamic University, Kushtia, Bangladesh
Email: asad@math.iu.ac.bd